- Analisis & perancangan sistem ⇒ Metode u/ mengembangkan dan memelihara sistem yang melakukan fungsi bisnis dasar
  - ↳ Analisis : Pembelajaran komponen dan model
  - ↳ Perancangan : Perancangan keseluruhan perangkat lunak
- Metodologi ⇒ kumpulan prosedur, teknik, dan tools, yang didukung dengan filosofi
  - ↳ Task → Technique ⇒ Method → Methodology
  - ↳ Menstandardisasi proses, improve, belajar

Waterfall Model

Komunikasi — Rekayasa kebutuhan → Analisis kebutuhan → Perancangan Umum / rinci → Pengkodean dan Pengujian → Deployment → Pemeliharaan

Perencanaan — Pemodelan — Konstruksi — Deployment

Sekuensial
Terdefinisi dengan baik
Safety

Top-Down Model

↳ by IBM 1970-an
↳ kebutuhan tingkat tinggi → rendah
↳ Contohnya design menu.

Model Pengembangan
- → Top-down model
- → Bottom-top model
- → Hybrid model

Bottom-top model

↳ Kebutuhan tingkat rendah → tinggi
↳ butuh ~~eb~~ koordinasi intense
↳ ~~Mod~~ Modular system
↳ Reusable component.

Paradigma Prototyping

→ Rencana Singkat → Pemodelan Rancangan Singkat → Pengembangan Prototype → development, delivery, feedback → Komunikasi → Rencana Singkat

Hybrid model

↳ ~~Desig~~ Design team: Top-bottom model
↳ Development team: Bottom-top model

Prototype ⇒ Supaya ada gambaran.

Pentingnya testbed yang tepat.

RAD ⇒ Rapid Application Development.
↳ Pengembangan aplikasi dalam proses yang singkat

Business modeling → Data modeling → Process modeling → Application generation → Testing and turnover

Spiral Model

Planning → Risk analysis → Engineering → Construction & release → Customer evaluation
↳ Task region

Concept dev → New product → Product enhancement → Product maintenance

Project Entry Point Axis

Boehm's win win Spiral Model

1. Identify next-level stakeholders
2. Identify stakeholder's win conditions
3. a) Reconcile win conditions
   b) Establish next-level objectives, constraints and alternatives
4. Evaluate process and product alternatives and resolve risks.
5. Define next level of product and process, including partitions
6. Validate product and process definitions
7. Review and comment

Specialized Process Model

Component based development ⇒ "Reusable component concept."
The formal methods model ⇒ "Use formal mathematical specification"
Aspect oriented software development ⇒ "focus on ~~cost~~ several aspects defined by user."

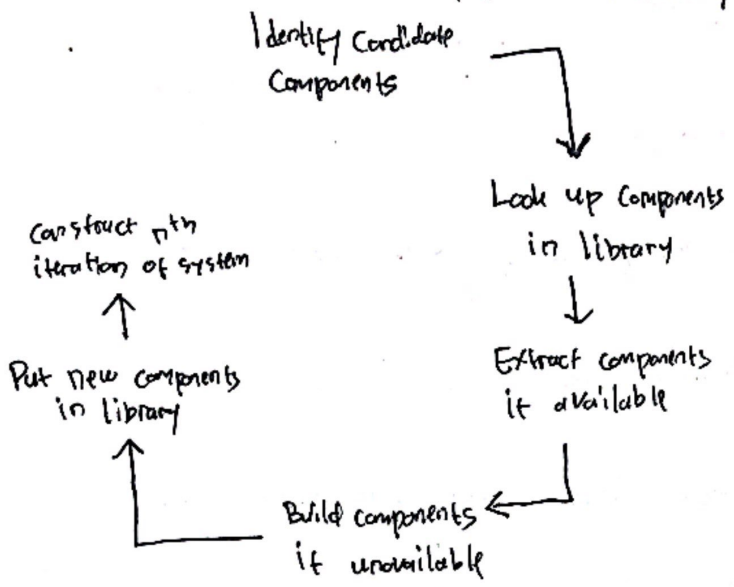
Coding → Unit testing → Integration testing → User acceptance testing

Agile Methods

Agile modeling
Agile unified process (AUP)
Dynamic systems development method (DSDM)
Extreme programming
Scrum

Pengembangan berbasis komponen

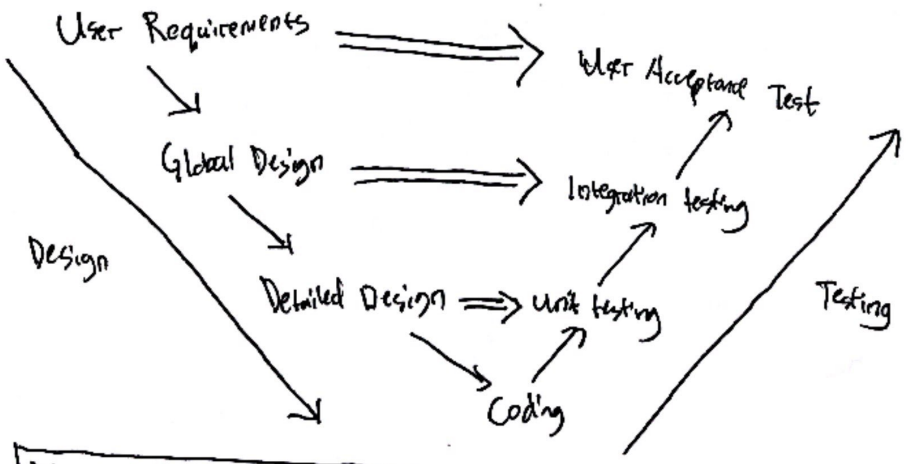
- → kombinasi metode spiral
- → komponen disimpan didalam repositori
- → Prinsip "reusability".



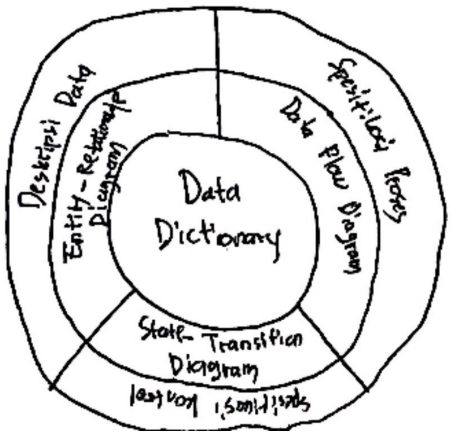
Metode Formal

- → Matematis formal dari aktivitas software
- → Menentukan, mengembangkan, dan memverifikasi sistem

V-Model

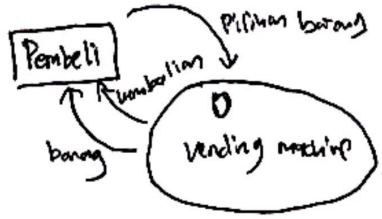


Model Analisis Terstruktur



(DFD)

Diagram konteks → Konteks diagram menjelaskan fungsi sistem dan mendefinisikan jangkauan dari sistem



Arsitektur dipartisi agar...

- Software akan lebih mudah diuji
- Software akan lebih mudah dirawat ('maintain')
- Efek propagasi kesalahan berkurang
- Software lebih mudah ditambah kurang modulnya.

Partisi arsitektur → Horizontal
→ Vertical

Abstraksi ⇒ ~~Cara~~ Suatu cara u/ mengatur kompleksitas pada suatu sistem komputer.
↳ # Abstraksi data
Abstraksi prosedural | tindakan | action)

Struktur data ⇒ Representasi dari <u>hubungan logis</u> antar elemen-elemen data.

Struktur prosedur ⇒ Abstraksi kendali.

Modularisasi dilakukan sampai mencapai titik optimum antara jumlah modul dengan biaya integrasi.